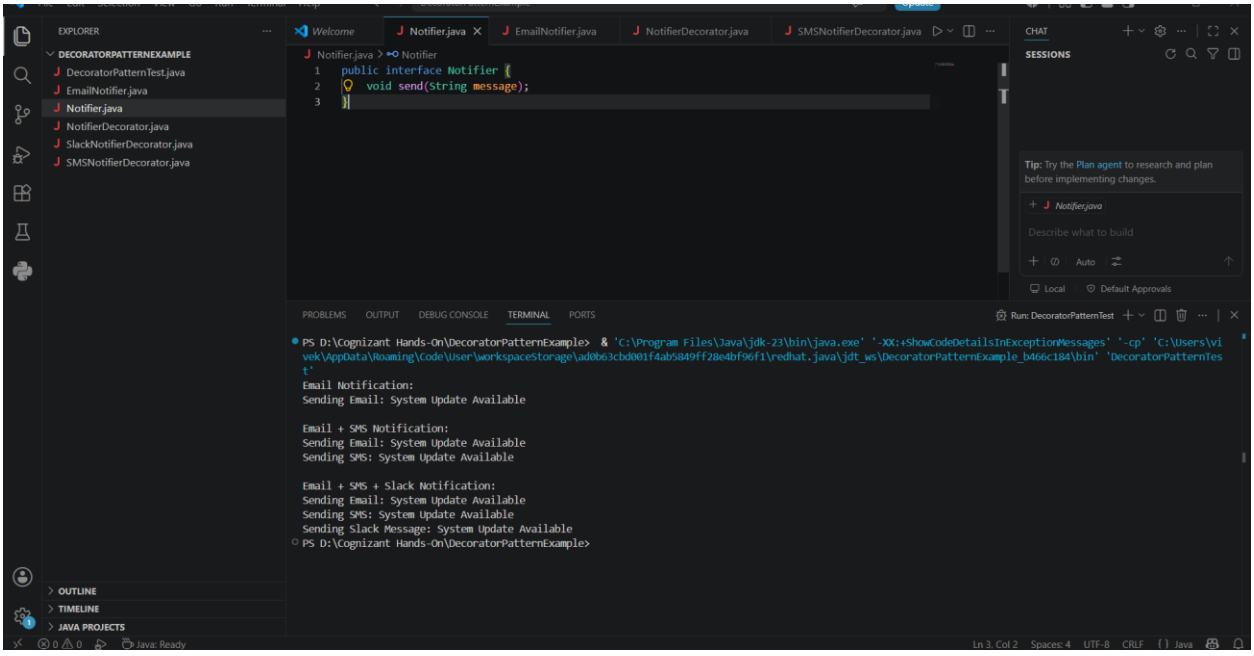
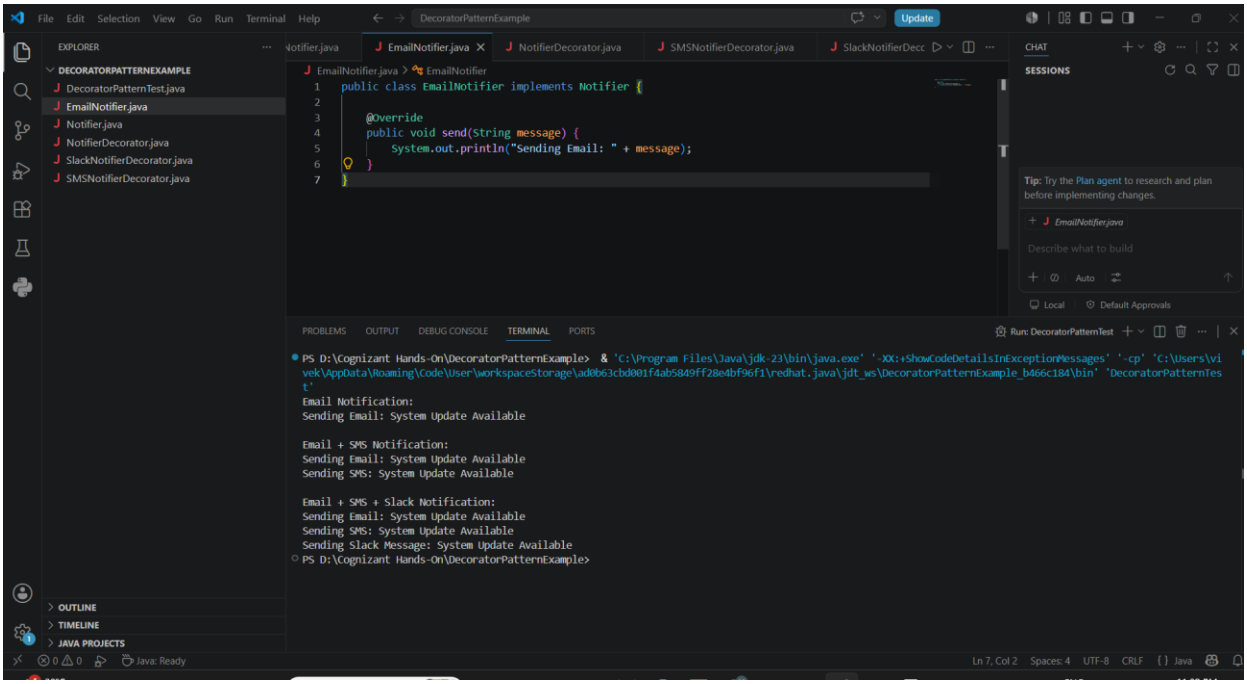
Hands-On

Exercise 5: Implementing the Decorator Pattern

As part of this module, I successfully implemented, executed, and tested the Decorator Pattern in Java by developing a project named DecoratorPatternExample. I created a Notifier interface with a send() method and implemented an EmailNotifier class as the base notification component. To add additional notification channels dynamically, I developed an abstract NotifierDecorator class and concrete decorators such as SMSNotifierDecorator and SlackNotifierDecorator. After completing the implementation, I executed the program and tested different combinations of notification channels to verify that new functionalities could be added without modifying the existing code. This hands-on exercise helped me understand how the Decorator Pattern enhances flexibility by dynamically extending the behavior of objects while maintaining a clean and maintainable code structure.

Coding:




File Edit Selection View Go Run Terminal Help
DecoratorPatternExample
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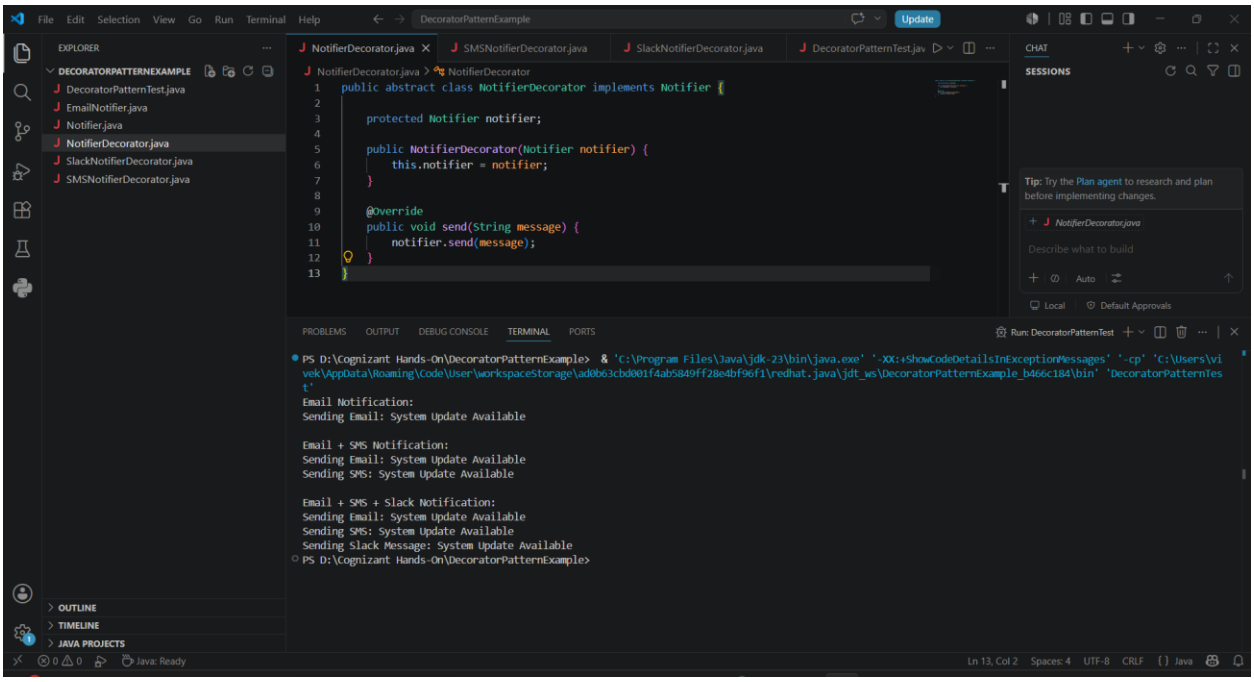
DECORATORPATTERNEXAMPLE
DecoratorPatternTest.java
EmailNotifier.java
Notifier.java
NotifierDecorator.java
SlackNotifierDecorator.java
SMSNotifierDecorator.java
EmailNotifier.java
NotifierDecorator.java
SMSNotifierDecorator.java
SlackNotifierDecc
EmailNotifier.java > EmailNotifier
1 public class EmailNotifier implements Notifier {
2
3 @Override
4 public void send(String message) {
5 System.out.println("Sending Email: " + message);
6 }
7 }
CHAT
SESSIONS
Tip: Try the Plan agent to research and plan before implementing changes.
EmailNotifier.java
Describe what to build
Auto
Local
Default Approvals
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Run: DecoratorPatternTest
PS D:\Cognizant Hands-On\DecoratorPatternExample> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\vivek\AppData\Roaming\Code\User\workspaceStorage\ad0b63cbd001f4ab5849ff28e4bf96f1\redhat.java\jdt_ws\DecoratorPatternExample_b466c184\bin' 'DecoratorPatternTest'
Email Notification:
Sending Email: System Update Available
Email + SMS Notification:
Sending Email: System Update Available
Sending SMS: System Update Available
Email + SMS + Slack Notification:
Sending Email: System Update Available
Sending SMS: System Update Available
Sending Slack Message: System Update Available
PS D:\Cognizant Hands-On\DecoratorPatternExample>
OUTLINE
TIMELINE
JAVA PROJECTS
Java: Ready
Ln 7, Col 2
Spaces: 4
UTF-8
CRLF
Java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SlackNotifierDecorator.java
DecoratorPatternTest.jav
NotifierDecorator.java > NotifierDecorator
1 public abstract class NotifierDecorator implements Notifier {
2
3 protected Notifier notifier;
4
5 public NotifierDecorator(Notifier notifier) {
6 this.notifier = notifier;
7 }
8
9 @Override
10 public void send(String message) {
11 notifier.send(message);
12 }
13 }
CHAT
SESSIONS
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NotifierDecorator.java
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Email Notification:
Sending Email: System Update Available
Email + SMS Notification:
Sending Email: System Update Available
Sending SMS: System Update Available
Email + SMS + Slack Notification:
Sending Email: System Update Available
Sending SMS: System Update Available
Sending Slack Message: System Update Available
PS D:\Cognizant Hands-On\DecoratorPatternExample>
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Java: Ready
Ln 13, Col 2
Spaces: 4
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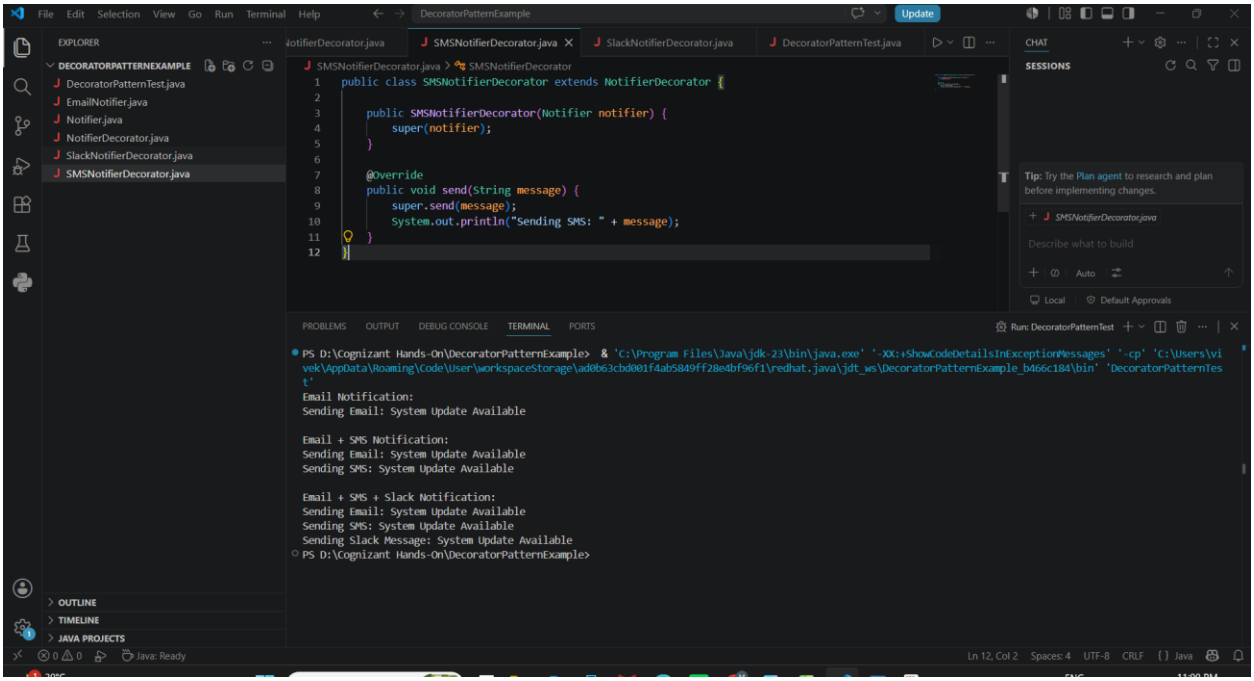
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SlackNotifierDecorator.java
DecoratorPatternTest.java
CHAT
SESSIONS
public class SMSNotifierDecorator extends NotifierDecorator {
public SMSNotifierDecorator(Notifier notifier) {
super(notifier);
}
@Override
public void send(String message) {
super.send(message);
System.out.println("Sending SMS: " + message);
}
}
Tip: Try the Plan agent to research and plan before implementing changes.
SMSNotifierDecorator.java
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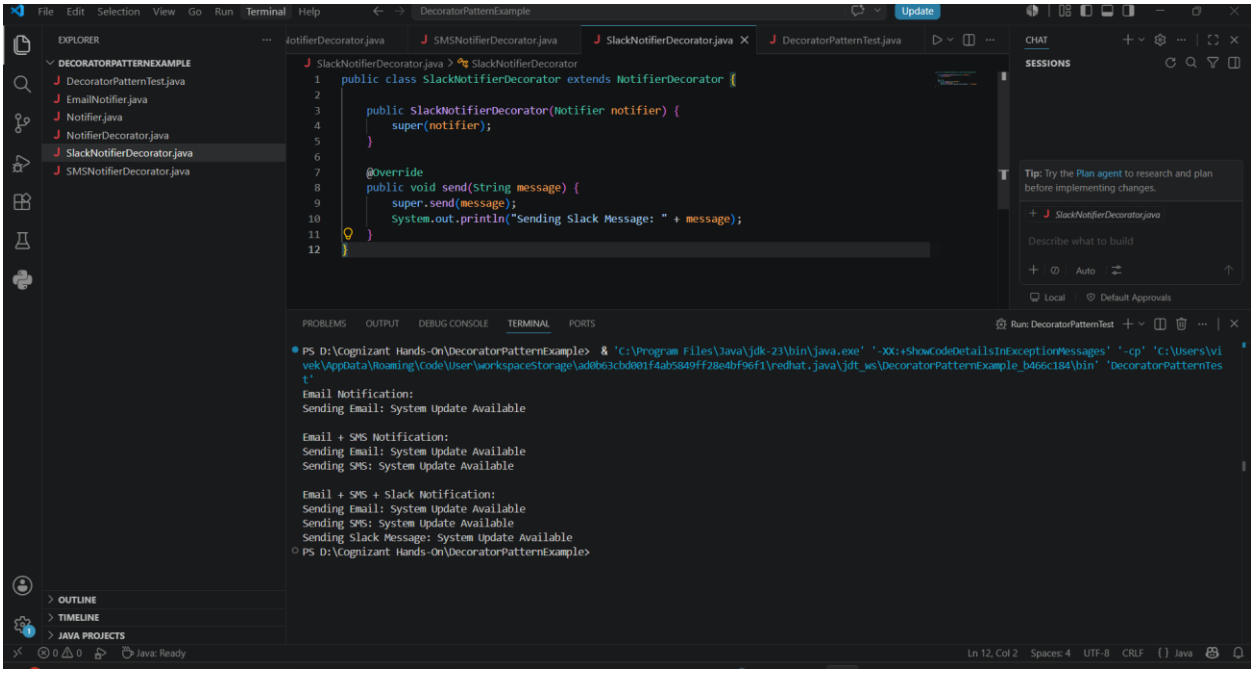
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SlackNotifierDecorator.java
DecoratorPatternTest.java
CHAT
SESSIONS
public class SlackNotifierDecorator extends NotifierDecorator {
public SlackNotifierDecorator(Notifier notifier) {
super(notifier);
}
@Override
public void send(String message) {
super.send(message);
System.out.println("Sending Slack Message: " + message);
}
}
Tip: Try the Plan agent to research and plan before implementing changes.
SlackNotifierDecorator.java
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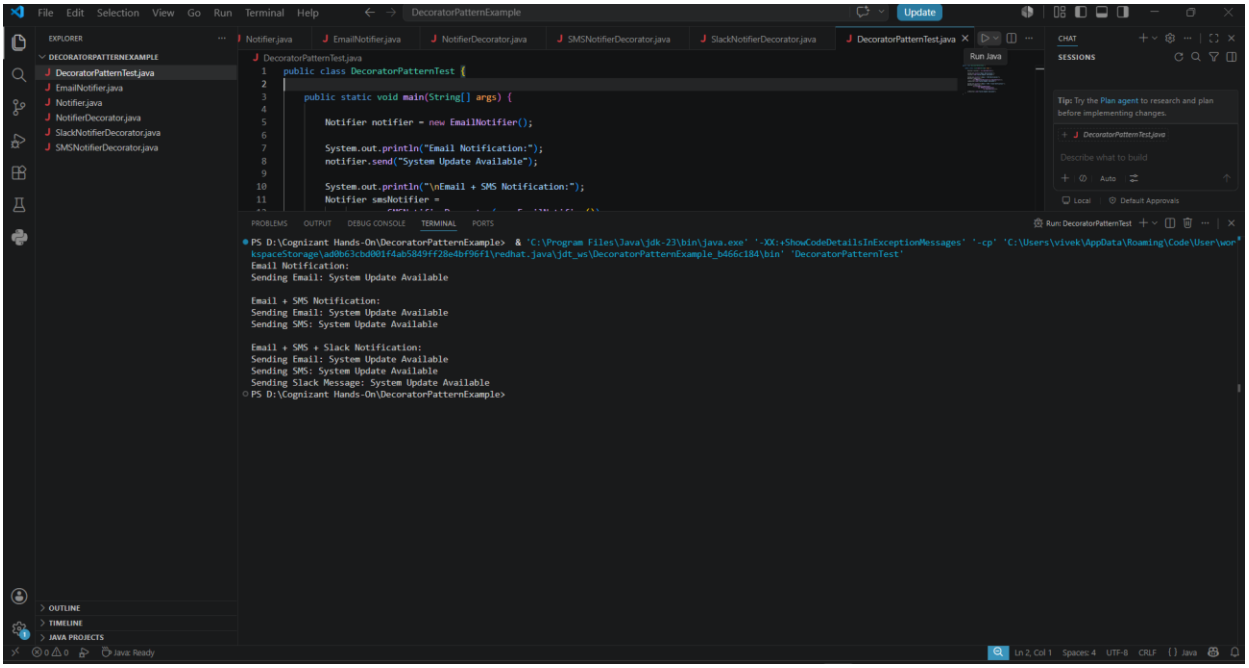
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Notifier.java
EmailNotifier.java
NotifierDecorator.java
SMSNotifierDecorator.java
SlackNotifierDecorator.java
DecoratorPatternTest.java
Run Java
public class DecoratorPatternTest {
public static void main(String[] args) {
Notifier notifier = new EmailNotifier();
System.out.println("Email Notification:");
notifier.send("System Update Available");
System.out.println("\nEmail + SMS Notification:");
Notifier smsNotifier =
CHAT
SESSIONS
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